

The WITS Geeks

Architecture Document

introduction

The architecture diagram illustrates the separation of concerns in the SA Learnerships platform. It shows how the user interface, server-side logic, data storage, and external authentication work together through well-defined interfaces.

Architecture Layers

- **Presentation Layer** – The browser runs the frontend, built with HTML, CSS, JavaScript, and Chart.js. Static files are served by Firebase Hosting, making the application fast and globally accessible.
- **Business Logic Layer** – The Express API, hosted on Render, handles all business logic. It includes route handlers, controllers, JWT authentication middleware, the analytics module, and standardised reference data (qualifications, skills, institutions, and locations) loaded at startup.
- **Data Layer** – MongoDB Atlas stores all persistent data across four collections: users, opportunities, applications, and notifications. The server uses Mongoose to query and update the database.
- **External Service** – Google OAuth 2.0 provides third-party authentication. The browser initiates an OAuth login, and the callback is returned to the backend.

All API requests from the browser to the backend are secured with HTTP-only JWT cookies. Controllers interact with the database through Mongoose, and the frontend communicates with Firebase only for static assets.

